Cheatsheet AZ-305 — 09 · Azure SQL, Cosmos DB and data platforms

1. Azure SQL products

Product	What it is	Choose when	Rule out when
Single Database	One isolated DB on a logical server	Independent DB; **Serverless**; modern features	Cross-database queries, three-part naming, SQL Agent
Elastic Pool	Several single DBs sharing compute	Many DBs with variable usage; **reserved capacity**; Hybrid Benefit; autoscaling across DBs	They ask for Serverless (does not exist in a pool)
Managed Instance	Instance almost 100 % compatible with SQL Server	**Cross-database**, linked servers, SQL Agent, three-part naming, **server-side transactions**, lift-and-shift migration	They ask for Serverless or the minimum cost without instance features
SQL Server on VM	IaaS	OS control, full compatibility, features MI does not cover	"minimize administrative effort"

- Two DBs + "server-side transactions across DB1 and DB2" + "minimize admin effort" → **MI**.
- "server.database.table" → **MI or VM** (two correct); never Single DB.
- "automatically scales compute + per-second billing" → **Single DB, General Purpose Serverless**.
- "20 DBs, varying usage, 99.99 %, dynamic compute, **reserved capacity**" → **Elastic pool** (Serverless does not support reservations).
- "License Mobility / Software Assurance" → **Azure Hybrid Benefit** → **vCore** model (pool or MI).

2. Purchasing models

	DTU	vCore
Unit	CPU+RAM+IO bundle (DTU)	vCores + RAM + storage separately
Tiers	Basic → Standard → Premium	General Purpose → Business Critical → Hyperscale
Hybrid Benefit	✗	✓
Reserved capacity	✗	✓
Serverless	✗	✓ (GP only, single DB)
Managed Instance	✗	✓
Status	Legacy, small DBs	Recommended

3. Tiers (DTU ↔ vCore translation)

Tier	Model	Equivalent to	Storage	I/O latency	HA	Zone-redundant	Max size	Serverless
Basic	DTU	—	Remote	High	Basic	✗	**2 GB**	✗
Standard	DTU	General Purpose	Remote	Medium	Compute reassignment	✗	1 TB	✗
General Purpose	vCore	Standard	Remote (Premium Storage)	Medium	Compute reassignment	✓ optional	4 TB	✓
Premium	DTU	Business Critical	**Local SSD**	**Minimum**	Primary + **synchronous** secondaries	✓	4 TB	✗
Business Critical	vCore	Premium	**Local SSD**	**Minimum**	Primary + 3 synchronous secondaries	✓ (survives 2 zones)	4 TB	✗
Hyperscale	vCore	—	Page servers (decoupled)	Low-medium	Optional HA replicas, **named replicas** for reads	✓	**100 TB**	✗

- "minimize I/O latency" + "2 zones fail" + "automatic failover" → **Business Critical** (or Premium in DTU).
- "failover without data loss" + "zone outage" + "min cost" (DTU options) → **Premium** (Standard has no zone redundancy).
- "expand to 50 GB, minimize cost" → **Standard**, not Premium.
- "> 4 TB" / "50 TB relational" / "200-300 concurrent reads" → **Hyperscale** (Synapse dedicated pool = **128 concurrent queries** max).
- "improve performance, minimize admin effort" → **Automatic tuning** (acts) vs Advisor/QPI/Monitor (only report).

4. Security in Azure SQL

Need	Solution
File encryption at rest	**TDE** (with CMK in Key Vault if "manage own keys"; Policy DeployIfNotExists to enable it)
Column never in clear text, only the client decrypts	**Always Encrypted**
Some see everything, others partially	**Dynamic Data Masking**
Firewall that travels with the failover	**Database-level** rules
Entra ID users without keys	Entra ID authentication + managed identity
Auditing / threats	SQL Auditing, Defender for SQL

5. Cosmos DB

Topic	Rule
APIs	**NoSQL** (JSON documents, SQL-like queries, **non-relational**) · **PostgreSQL** (relational, standard SQL, distributed, geo-replicas) · MongoDB · Cassandra (wide-column, CQL) · Gremlin (graphs) · Table
"store and access relationally" + "SQL queries"	**PostgreSQL** (NoSQL fails on "relational")
Multi-region writes	Only service with local writes in several regions; "read and write in local region" → Cosmos DB
Consistency levels	Strong · Bounded staleness · Session · Consistent prefix · Eventual
Partition key	**Immutable**; changing it = **create a new container** and migrate
Data RBAC	Built-in Data Reader / Contributor with Entra ID identities, no account keys or resource tokens
Analytics without ETL	**Synapse Link** → columnar analytical store, near real time, without consuming RUs
RU alerts	Log Analytics query over requestCharge_s
Catalog with variable schema + global distribution	Cosmos DB

6. Other data platforms

Service	What for	Signal
Synapse dedicated SQL pool	MPP warehouse for heavy analytical queries	"data warehouse" with **≤128 concurrent**
Synapse serverless SQL	Query files in the lake without loading	"query data directly in the lake"
Data Lake Storage Gen2	Lake with directories by date (Event Hubs Capture), POSIX ACLs	"organize by date/time", "Hadoop"
Databricks	Spark; **Credential Passthrough** (user identity → ACLs) or Unity Catalog; on-prem app auth with **service principal**	"user identity for folder ACLs"
Data Factory / Synapse pipelines	Low-code ETL/ELT; **Self-hosted IR** for on-prem; **Azure-SSIS IR** for SSIS packages (SSISDB in **Azure SQL Database**)	"transform during load", "SSIS packages"
Data Share	Share snapshots with another organization	"partner", "restricted access to snapshots"
Purview	Catalog, classification, lineage	"discover sensitive data"
Azure Database for MySQL/PostgreSQL Flexible	Regional DR with **promoted read replica** (min downtime); geo-backup = geo-restore (hours)	"minimize downtime to paired region" → read replica
Table Storage	Independent tables over REST, minimum cost, GRS	"REST + independent tables + second region + min cost"
Azure Cache for Redis	Session state / cache next to the app	"session state", "serializable"

7. Decision tree (Azure SQL)

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Instance or DB?
├─ Cross-database / Agent / linked servers ..... Managed Instance (VM if you need the OS)
├─ Many DBs with variable usage / reservations . Elastic pool
└─ Isolated DB
    ├─ Autoscaling + per second ................ General Purpose Serverless
    ├─ Minimum latency / 2 zones / sync ........ Business Critical (Premium in DTU)
    ├─ > 4 TB / high concurrency ............... Hyperscale
    ├─ < 2 GB, minimum cost, no HA ............. Basic
    └─ General ................................. General Purpose (Standard in DTU)
```

8. Signals → answer

Wording	Answer
"server-side transactions across DB1 and DB2" + min effort	Managed Instance
"server, database, and table name"	MI + SQL on VM
"per second billing"	Single DB, General Purpose Serverless
"reserved capacity" + many DBs	Elastic pool
"Software Assurance"	vCore + Hybrid Benefit
"minimize I/O latency" + "two zones"	Business Critical
"50 TB relational" + "200-300 concurrent reads"	Hyperscale
"read and write in the local region"	Cosmos DB (multi-region writes)
"relationally" + Cosmos API	PostgreSQL
"change partition key, first step"	Create a new container
"Entra users read access to Cosmos data, no keys"	Built-in Data Reader (RBAC)
"near real-time analytics on Cosmos DB, no ETL"	Synapse Link
"SSISDB catalog, no hybrid connectivity, min cost"	Azure SQL Database
"MySQL flexible, minimize downtime to paired region"	Read replica

9. Mistakes to avoid

1. Serverless in an elastic pool or MI → does not exist.
2. Reserved capacity with Serverless → incompatible.
3. Standard (DTU) for zone redundancy → it does not have it.
4. Synapse dedicated pool with >128 concurrent reads → Hyperscale.
5. Cosmos NoSQL as "relational" because it has SQL queries.
6. Premium for 50 GB when Standard is enough and they ask for minimum cost.
7. Geo-restore/geo-backup when they ask to minimize downtime.